3 **(a)** State what is meant by *simple harmonic motion*.

……

……

…… [2]

(b) A smooth ball of mass *m* is held between two fixed points A and B by two similar springs, each of spring constant *k*, as shown in Fig. 3.1.

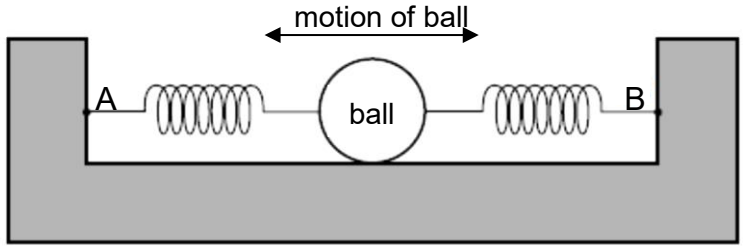

Fig. 3.1

When the ball is in equilibrium, the extension of each spring is *e*. The ball is then displaced a small distance *x* to the right along the axis of the springs. The ball is then released and oscillates on the smooth surface along the straight line joining points A and B.

(i) Show that the acceleration *a* of the ball is given by the equation

$$a = -\frac{2kx}{m}$$

[3]

(ii) The mass *m* of the ball is 900 g and the spring constant *k* is 120 N m^{-1}. The maximum acceleration of the ball is 5.2 m s^{-2}.

1. Determine the frequency of oscillation of the ball.

frequency = …… Hz [2]

2. Determine the amplitude of the oscillation.

amplitude = …… m [2]

3. Determine the maximum kinetic energy of the ball.

maximum kinetic energy = …….. J [2]